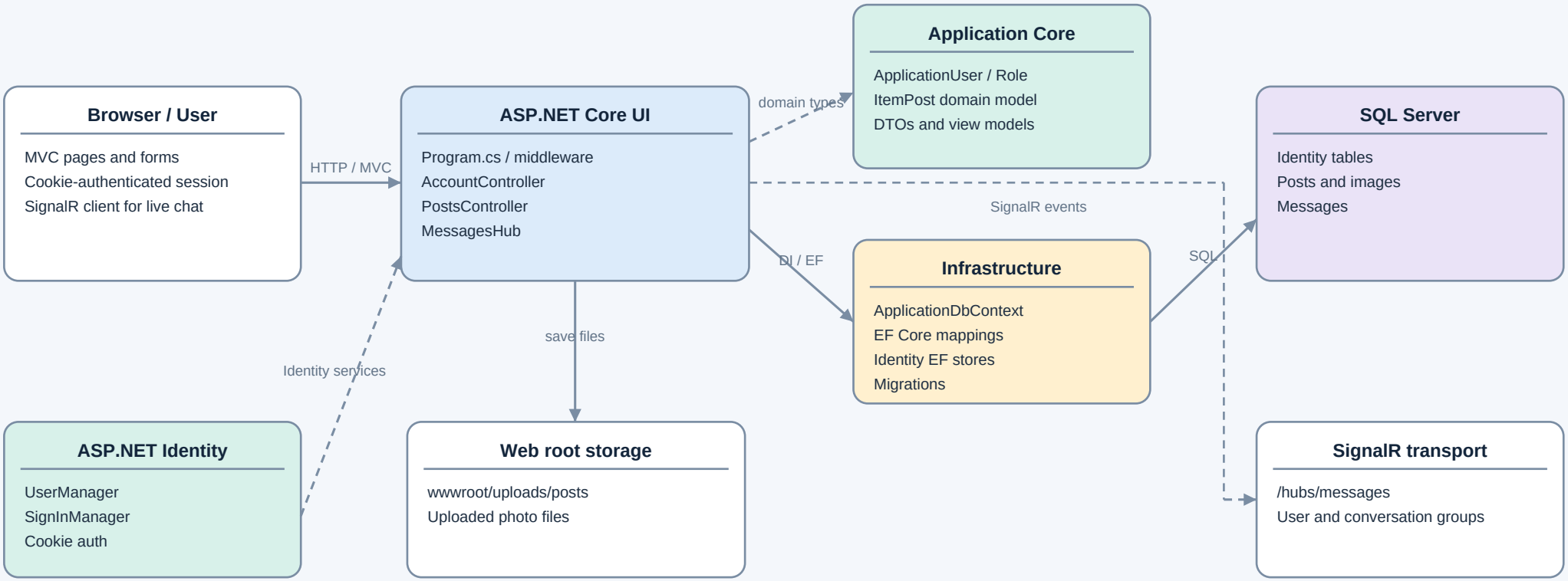
System architecture

Component diagram



Request path

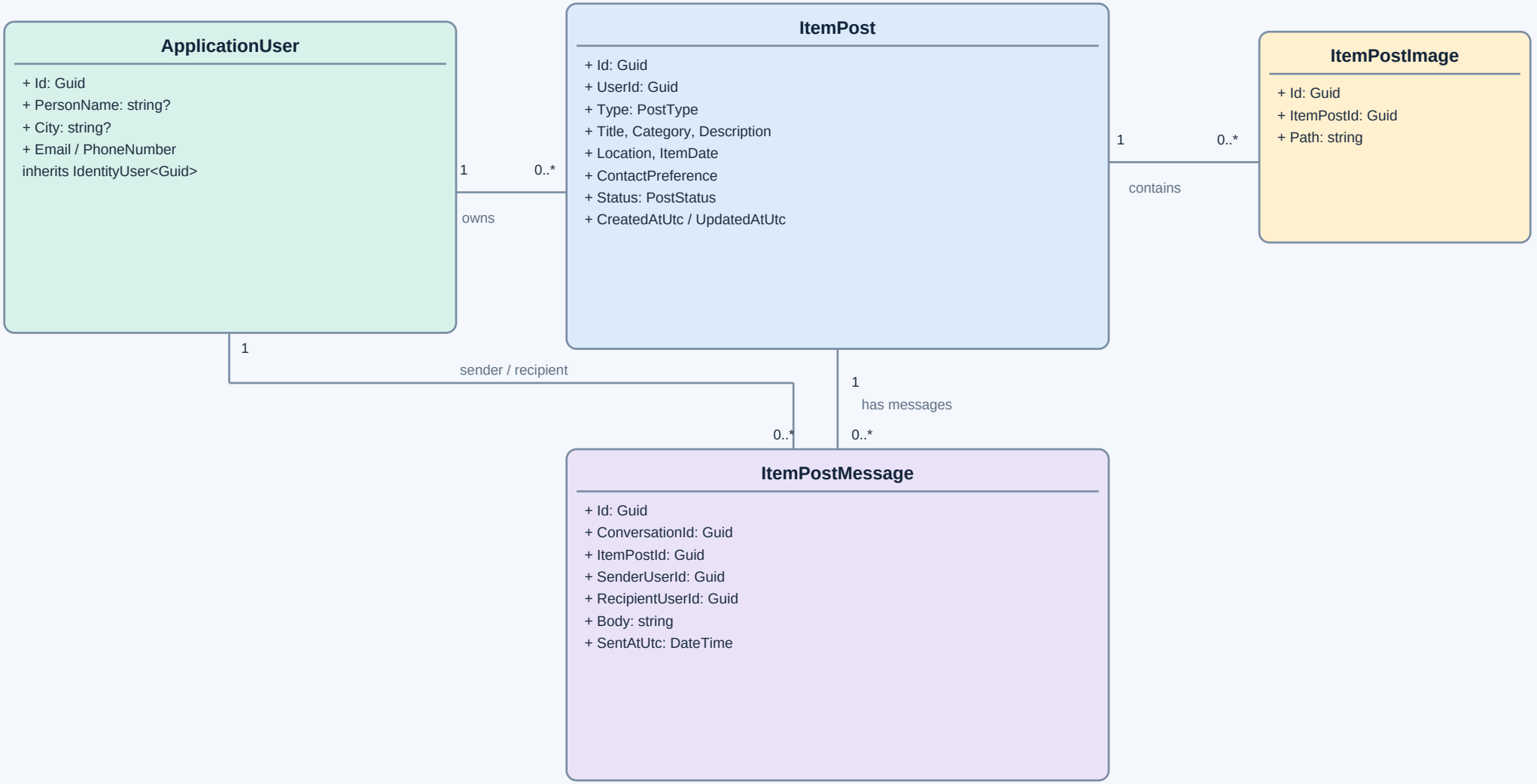
MVC requests pass through routing, authentication, authorization, controllers, and EF Core. The app runs pending EF migrations at startup.

Persistence

SQL Server stores users, posts, image metadata, and messages; photo bytes are written to the web root. SignalR broadcasts persisted messages to participants.

Domain model

Class diagram - key entities and relationships



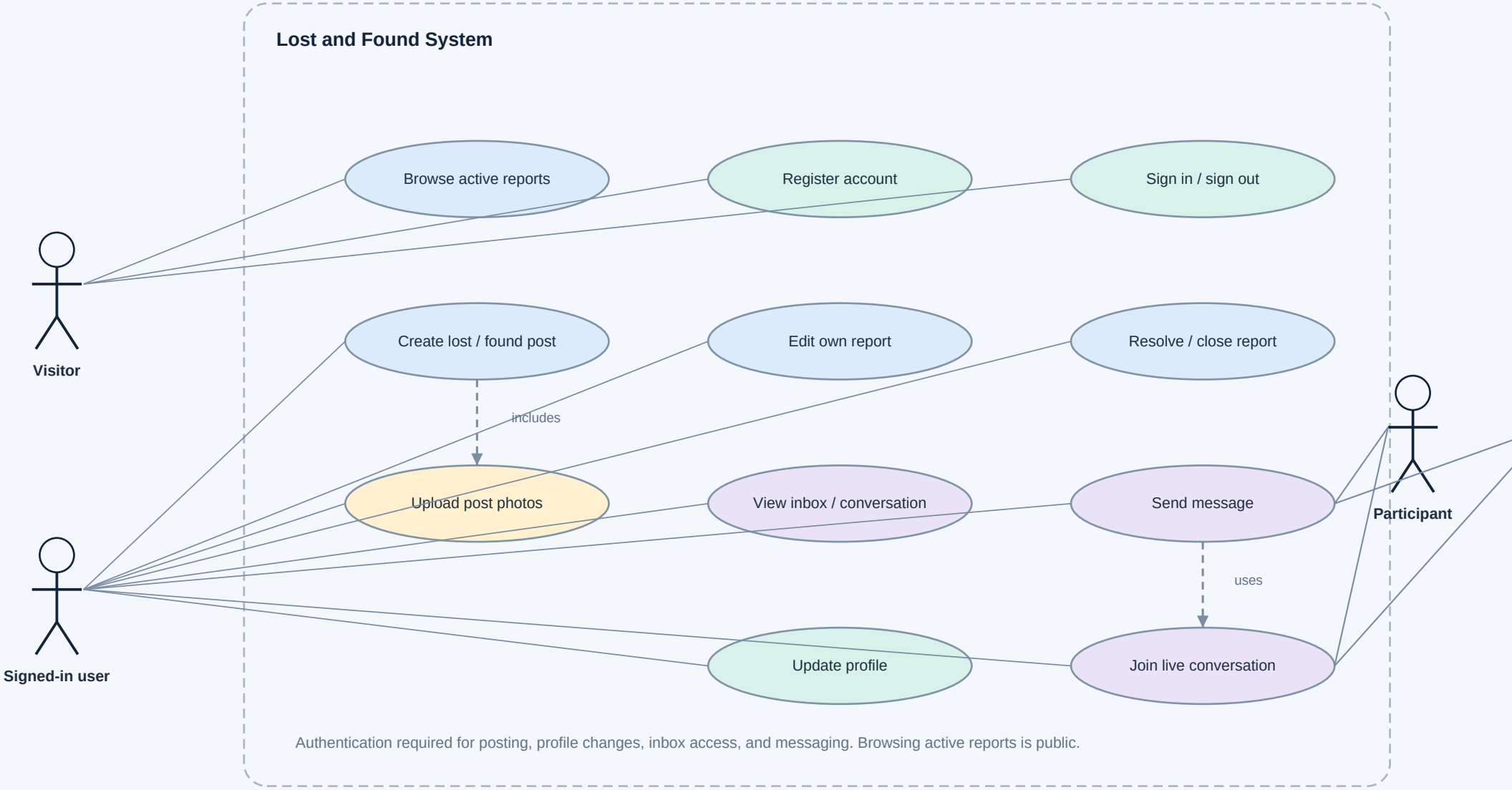
Enums

PostType = Lost | Found PostStatus = Active | Resolved | Closed ContactPreference = InAppMessagesOnly | ShowPhone

Database rules: a post belongs to one user; deleting a post cascades to its images and messages. User references are restricted from deletion.

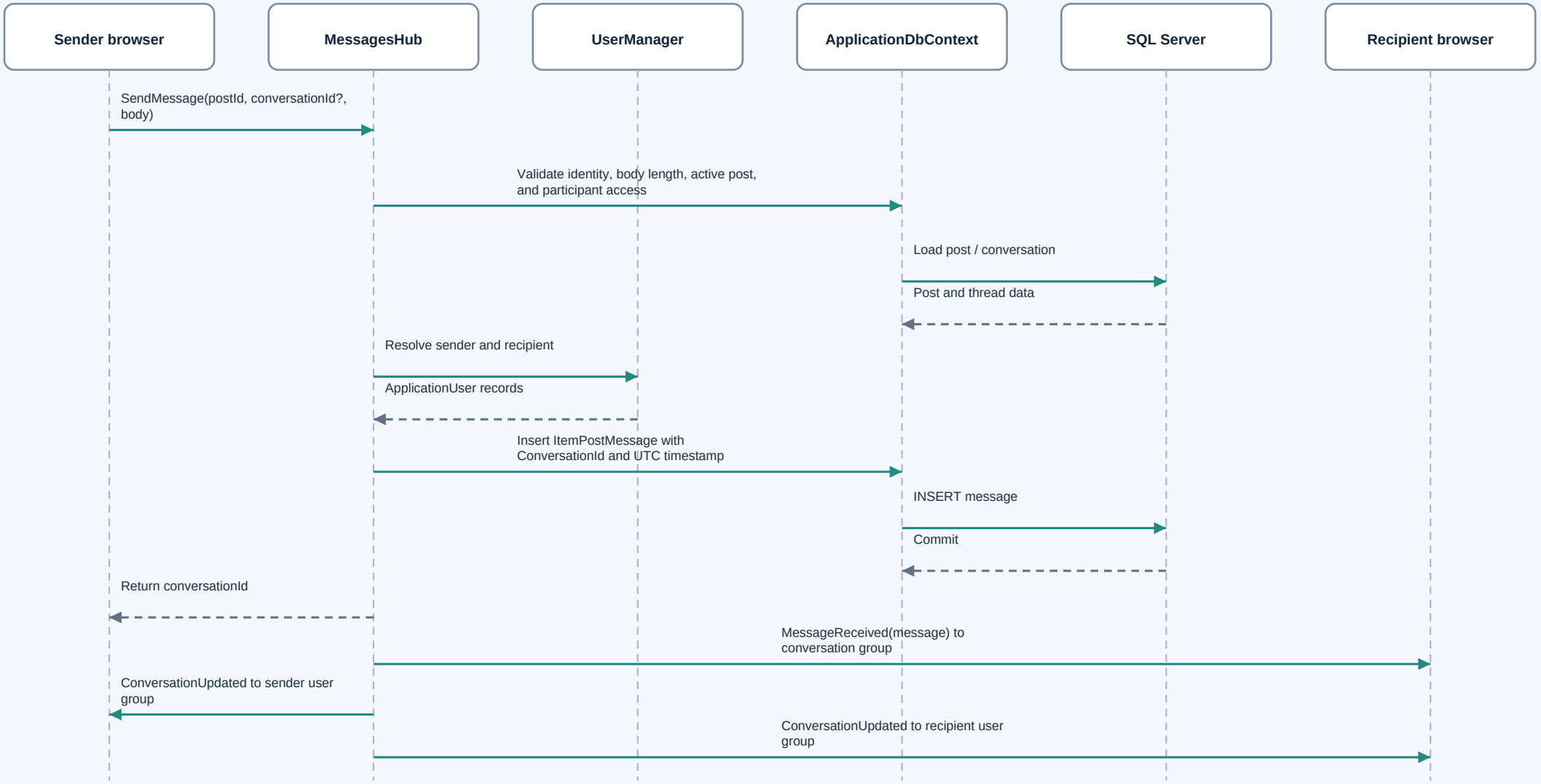
User capabilities

Use case diagram - primary actors and interactions



Live messaging

Sequence diagram - send a message over SignalR



A participant joins a conversation group only after the hub verifies they are a sender or recipient in that post's conversation.